1. Smart Surveillance System Selection

Requirements:

- Latency must be ≤ 100 ms.
- Accuracy must be ≥ 85%.
- Hardware supports only moderate computation.

Model	**Accuracy**	**Latency**	**Computation**	**Meets Requirements?**
Viola-Jones	78%	Low	Low	No
YOLO	92%	Moderate	Moderate	Yes
Deep Learning	95%	High	High	No

Comparison:
Viola-Jones has low latency and low computational requirements, but its accuracy is only 78%, which is below the required 85%. Deep Learning provides the highest accuracy of 95%, but its high latency and high computational requirement make it unsuitable for the given hardware. YOLO provides 92% accuracy with moderate latency and moderate computational requirements.

Selected Solution: YOLO

Justification:
YOLO is the best choice because it satisfies all three constraints: accuracy is above 85%, latency is within the acceptable limit, and the computational requirement is moderate. Therefore, YOLO provides a good balance between **accuracy, speed, and available hardware resources**, making it practical for city surveillance deployment.

2. Autonomous Vehicle Detection Logic

Requirements:

- Accuracy must be > 90%.
- Response time must be < 50 ms.
- Safety is the highest priority.

Method	Accuracy	Response Time	Meets Requirements?
Method A	88%	30 ms	No
Method B	93%	70 ms	No
Method C	91%	45 ms	Yes

Comparison:

Method A is fast with a response time of 30 ms, but its accuracy is only 88%, which does not satisfy the safety requirement. Method B provides 93% accuracy but takes 70 ms, exceeding the maximum response time. Method C provides 91% accuracy and responds within 45 ms.

Selected Solution: Method C

Justification:

Method C is the only method that satisfies **both accuracy and response-time constraints**. Its 91% accuracy is above 90%, while its 45 ms response time is below 50 ms. Therefore, Method C provides the best balance between **safety and real-time performance** and is the most practical choice for autonomous vehicles.

5. Depth Estimation Strategy

Requirements:

- Only one camera is available.
- Moderate depth accuracy is acceptable.

Technique	**Accuracy**	**Camera Requirement**	**Feasible?**
Stereo Vision	High	Two cameras	No
Structure from Motion	Moderate	Single camera	Yes

Comparison:

Stereo Vision can provide high depth accuracy, but it requires two cameras. This violates the given hardware constraint because only one camera is available. Structure from Motion can estimate depth using a single camera and provides moderate accuracy, which is acceptable for the application.

Selected Solution: Structure from Motion

Justification:

Structure from Motion is the most feasible solution because it works with a **single camera** and provides the required moderate depth accuracy. Although Stereo Vision may provide better accuracy, it cannot be deployed under the given hardware constraint. Therefore, Structure from Motion offers the best trade-off between **accuracy, hardware requirements, and cost**.

11. Traffic Surveillance Model Selection

Requirements:

- Accuracy $\geq 90\%$.
- Latency ≤ 80 ms.
- Must work under varying lighting conditions.

Method	**Accuracy**	**Latency**	**Meets Numerical Constraints?**
Method A	88%	60 ms	No
Method B	91%	85 ms	No
Method C	92%	75 ms	Yes

Comparison:

Method A has acceptable latency but its accuracy is below 90%. Method B achieves the required accuracy but has a latency of 85 ms, which exceeds the 80 ms limit. Method C achieves 92% accuracy and 75 ms latency, satisfying both numerical requirements.

Selected Solution: Method C

Justification:

Method C is the only option that satisfies both the **accuracy and latency constraints**. It provides 92% accuracy and operates at 75 ms latency. Since the question also requires operation under varying lighting, the final deployment should validate Method C under different lighting conditions before implementation. Based on the information provided, Method C is the most practical choice because it provides the best overall performance while remaining within the specified latency limit.

16. Autonomous Navigation Constraint

Requirements:

- Accuracy ≥ 95%.
- Latency < 70 ms.

Model	**Accuracy**	**Latency**	**Meets Requirements?**
Model A	94%	60 ms	❌ No
Model B	96%	80 ms	❌ No
Model C	95%	65 ms	✅ Yes

Comparison:

Model A has acceptable latency but its accuracy is only 94%, which is below the required 95%. Model B has 96% accuracy but its latency is 80 ms, exceeding the 70 ms limit. Model C provides exactly 95% accuracy and has a latency of 65 ms.

Selected Solution: Model C

Justification:

Model C is the only option that satisfies **both constraints**. Its accuracy is 95%, meeting the minimum requirement, and its latency is 65 ms, which is below the 70 ms limit. Although Model B has slightly higher accuracy, its latency is too high for autonomous navigation. Therefore, Model C provides the best balance between **accuracy and response speed** and is the most practical solution for deployment.